Chapter 2

Crunching the Numbers with Copilot

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In 2006, the British mathematician and data scientist Clive Humbly famously said, "Data is the new oil." Just as with oil, however, simply having data isn't of much use. The real power of data is in making sense of it through a process called *data analytics.* This is why Peter Sondergaard, the senior vice president of the technology research firm Gartner, Inc., said, "Information is the oil of the 21st century, and analytics is the combustion engine."

In the brief time since Humbly and Sondergaard came up with their famous quotes about the power of data, we've had an explosion in the power of artificial intelligence that has made data even more useful and powerful. Just to push the metaphor perhaps further than it should fly: If analytics is the combustion engine, then AI, in combination with data and analytics, is a rocket ship. And now, you have a copilot to help you guide that rocket ship.

Using Microsoft Copilot in Excel can improve your understanding of data as well as help you use Excel wisely (with which many of us struggle). In

this chapter, you explore how this AI assistant can help you work with data and boldly go where no one has gone before.

Launching Copilot in Excel

Subscribers to Microsoft 365 Copilot or Copilot Pro will see a Copilot button on the right side of the Home tab of Excel for Microsoft 365 as well as in the desktop version of Excel. Clicking this icon opens the Excel Copilot sidebar, shown in Figure 2-1.

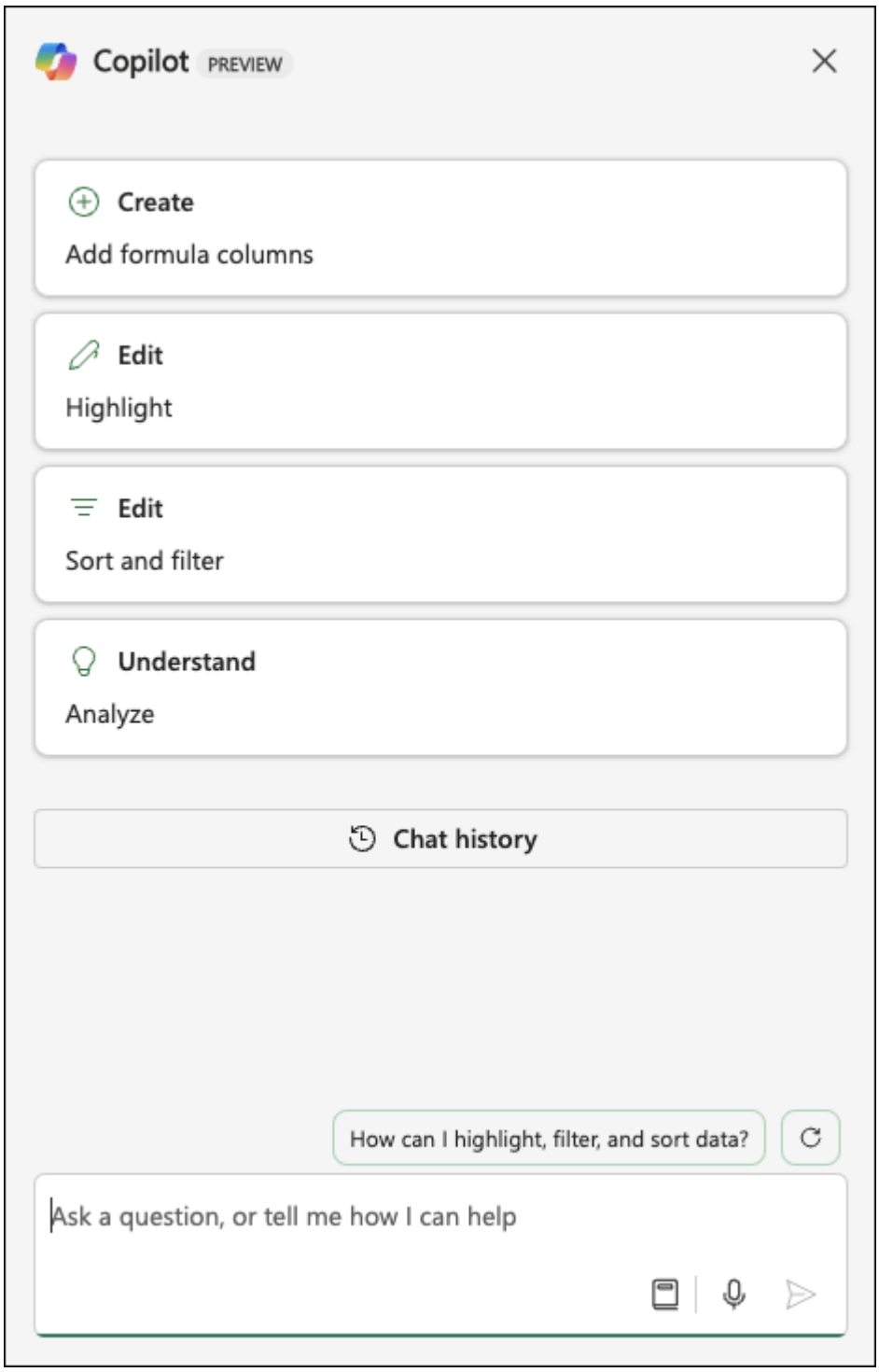


FIGURE 2-1: The Copilot for Excel sidebar.

At the top of the sidebar are buttons for the categories of special prompts that you can use in an Excel document:

- **Create.** Create-related prompts are prompts in which you ask Copilot to add new columns to your spreadsheet with formulas. To use a Create prompt, you can describe the column you want to add, and Copilot will attempt to do as you wish. You learn about Create prompts in the section of this chapter titled "Creating Formulas with Copilot's Assistance."
- **Edit.** Edit-related prompts assist you with highlighting, sorting, and filtering your data. The two categories of Edit-related prompts are Highlight and Sort and Filter.
- **Understand.** Understand-related prompts find and highlight interesting parts of your data, show insights (such as correlations and patterns) that Copilot finds in your data using charts and PivotTables, and can help you identify trends and patterns.

Below the special prompt buttons is the Chat History button, which takes you to a history of every prompt you've submitted in the current document.

Below the Chat History button is an area containing prompt suggestions. These prompt suggestions are context-aware, and they will change based on the content of your document and on your previous prompts in the chat.

At the bottom of the sidebar is the prompt input box, with some icons at the bottom for getting prompt suggestions and for using the microphone.

Talking about using Copilot in Excel without having some real data is only of limited use (like having a car without fuel!), so let's get some data and take Copilot for a spin.

Working with Data

One of the reasons the latest generation of AI is so good is that there's so much data available for AI to learn from. The internet has both fueled an

explosion of data collection and made it easier for anyone to access it. What and where is all this data?

Understanding the two kinds of data

People who work with AI and machine learning break data into two broad categories: structured and unstructured.

Unstructured data

Unstructured data includes websites, news articles, music, blogs, movies, television, and so much more. Unstructured data doesn't fit into neat rows and columns, but machine learning algorithms can still learn from it by looking for patterns and similarities between different pieces of data.

Unstructured data is used for tasks that seek to discover and recognize patterns. Examples of tasks in which AI learns from unstructured data include image recognition, natural language processing, speech recognition, and video analysis.

Structured data

Structured data includes data that can be organized into rows and columns. While movies and TV shows can't be "structured" data, you can create structured data *about* movies and TV shows. For example, you may create a list of the 100 best movies of all time and include data such as the year they were made, the director, the actors, the length of the movies, and so forth. Structured data also includes labeled data. Labeled data is data that may have previously been unstructured but that's been assigned labels to add context or meaning. An example of labeled data might include a collection of pictures that have been labeled as being either a picture of a cat or not a picture of a cat. Such a labeled collection of data can be used to train AI to identify pictures of cats.

The type of data that Excel works with is structured data.

Finding free data

You may have Excel files that you've created for work or to manage your personal finances. Or, you may have a list of important contacts, or even a list of passwords that you keep in an Excel file on your computer. All these files are likely only of interest to you and a small group of other people.

However, a universe of data is out there that's of interest to a much larger group of people. This includes weather data, data about political or charitable contributions, data about the stock market, and much more. Thanks to the internet, all of this data is readily available, often for free, to anyone who is interested in it and knows where to look.

Many sources of freely available data are available on the web. Organizations, universities, governments, data scientists, and enthusiasts regularly publish data and make it available for other people to study and analyze. Some of the best sources of free data include:

- `data.gov` **(**`https://data.gov`**).** `Data.gov` is the U.S. government's free and open source data repository. In particular, it contains a wealth of public economic data.
- **Google dataset search (**`https://datasetsearch.research.google.com`**).** If you know what you're looking for, this is a great place to start your search.
- **Kaggle (**`https://www.kaggle.com/datasets`**).** Kaggle is a community hub for data scientists and people who are interested in data science. Kaggle is a place to learn about AI and data science, collaborate with other data scientists, and even try your hand at data science challenges and competitions.
- **Datahub.io (**`https://datahub.io/collections`**).** Datahub contains mostly data related to business and finance. If you're looking for stock market data or information about property prices, you'll likely find it here.

For the first examples in this chapter, I downloaded a dataset from Kaggle titled "Tornados [1950 – 2022]." As you may have already guessed, this dataset contains information about every tornado in the United States between 1950 and 2022. If you want to download this dataset, you can do so at `https://www.kaggle.com/datasets/sujaykapadnis/tornados`.

Preparing the Data

After data has been collected, the next step in data analysis is to make sure it's properly formatted, that all columns have a consistent format, and that there isn't anything in the data that might throw off the results (such as missing data).

Since the tornados dataset was downloaded from a trusted source and has a high rating on `Kaggle.com`, I feel confident that the data itself is trustworthy. However, some things can be done to add some context to the data and make it more logical to Copilot.

Converting the data to a table

The downloaded tornado dataset is a comma-separated values (CSV) file, which can be easily opened using Excel. Copilot can't work with CSV files, however. To be able to use Copilot, you need to open the CSV file in Excel and then save it as the latest Excel format.

Once you've done that, you can click on the Copilot icon and ask Copilot to explain the data to you. On my first attempt, Copilot told me there was more data than it could work with and that I needed to convert the data to a table. After I figured out how to do that (by asking Copilot), I did the conversion to a table with these steps:

1. **Click in the data you want to convert to a table.**
2. **Click on Insert from Excel's *ribbon* (as the row of buttons across the top is called).**
3. **Click on Table in the Excel ribbon.**
4. **In the Create Table window that pops up, make sure that all the data in the current Excel spreadsheet is selected and that the checkbox next to My Table Has Headers is checked.**

The result of the preceding steps will be that your raw data will be converted into a nice-looking table with a header and alternating row formats, as shown in Figure 2-2.

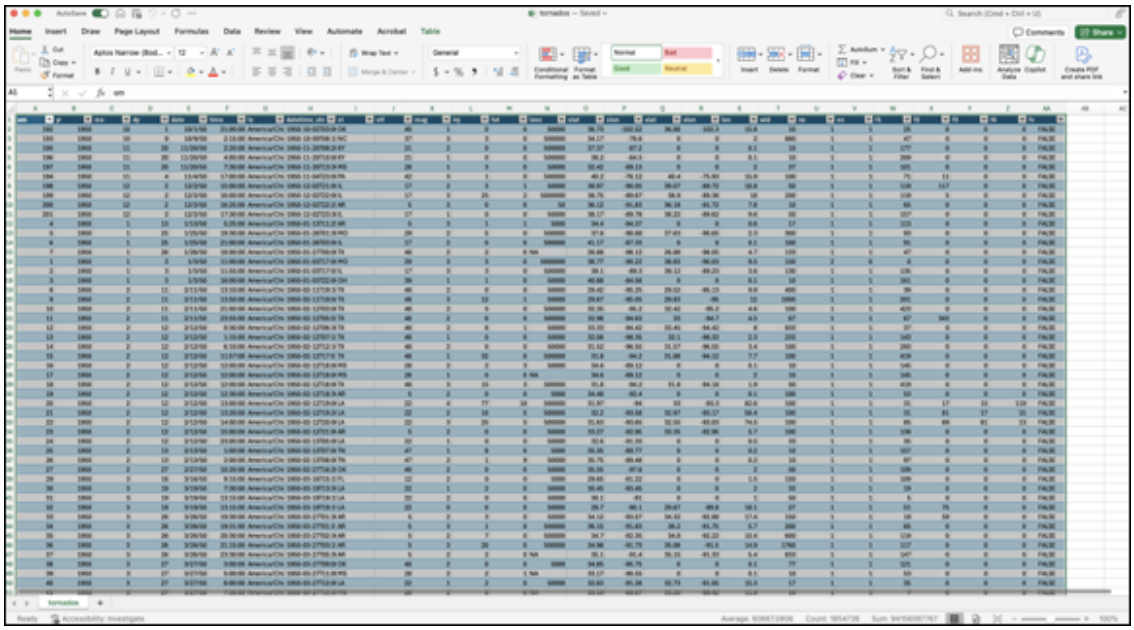


FIGURE 2-2: Creating a table from data.

Adding context to the headers

The column headers used in the spreadsheet are short and non-descriptive. It's possible that replacing some of the headers with plain English words could improve Copilot's ability to understand the data, and maybe Copilot can help us with that.



TIP You can find a key to the headers on the dataset's download page in the section called Data Card.

You can approach the problem of creating new headers for the Excel sheet in several ways without having to do it manually. One way is to ask Copilot in the browser to generate a single row table containing descriptive names for the columns.

My first attempt at using this approach caused Copilot to create a numbered list of descriptive names for the columns, but then when I asked it to create a CSV file containing the headers, it was able to. It did end up giving me a couple extra columns that aren't in the dataset, but those are easy enough to remove.

Figure 2-3 shows the result when I tried that approach.

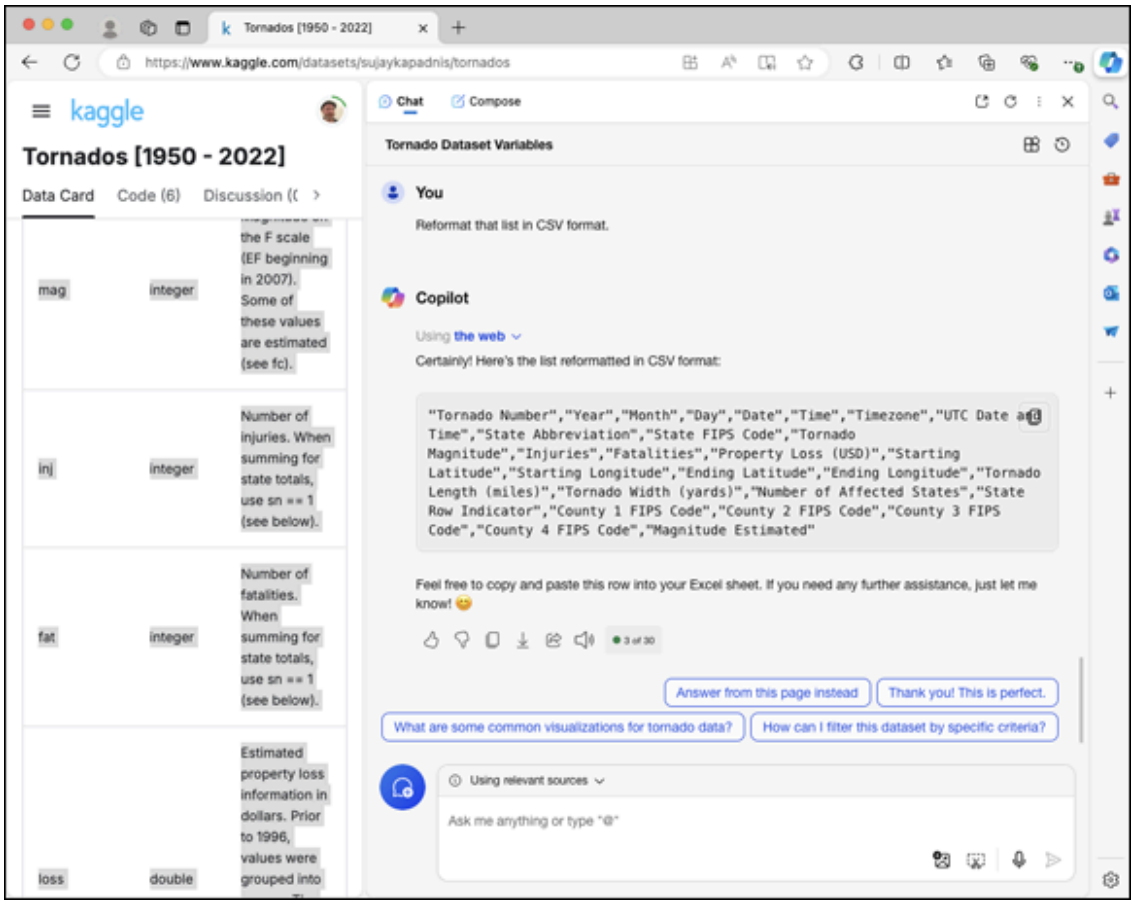


FIGURE 2-3: Asking Copilot in Edge to create a table header.

Another approach is to try pasting the table from the Data Card into Copilot in Excel and asking Copilot to make the changes directly in the Excel spreadsheet.

When I tried this approach, Copilot responded that "there is an issue with renaming the headers at the moment." I suspected that Copilot might not have the ability to rename the headers at all, so I tried an experiment: asking Copilot to just rename one header using the following prompt:

> *Please rename the tz column to Time Zone.*

This time, it came back with a message containing its proposed solution, and a button labeled Apply. Its proposed solution, however, was to replace every instance of the letters "tz" with "Time Zone" in the entire column, rather than just in the one cell that I asked it to modify. I pushed back, as shown in Figure 2-4.

I clicked the Apply button, and Copilot correctly changed the label of the time zone column.

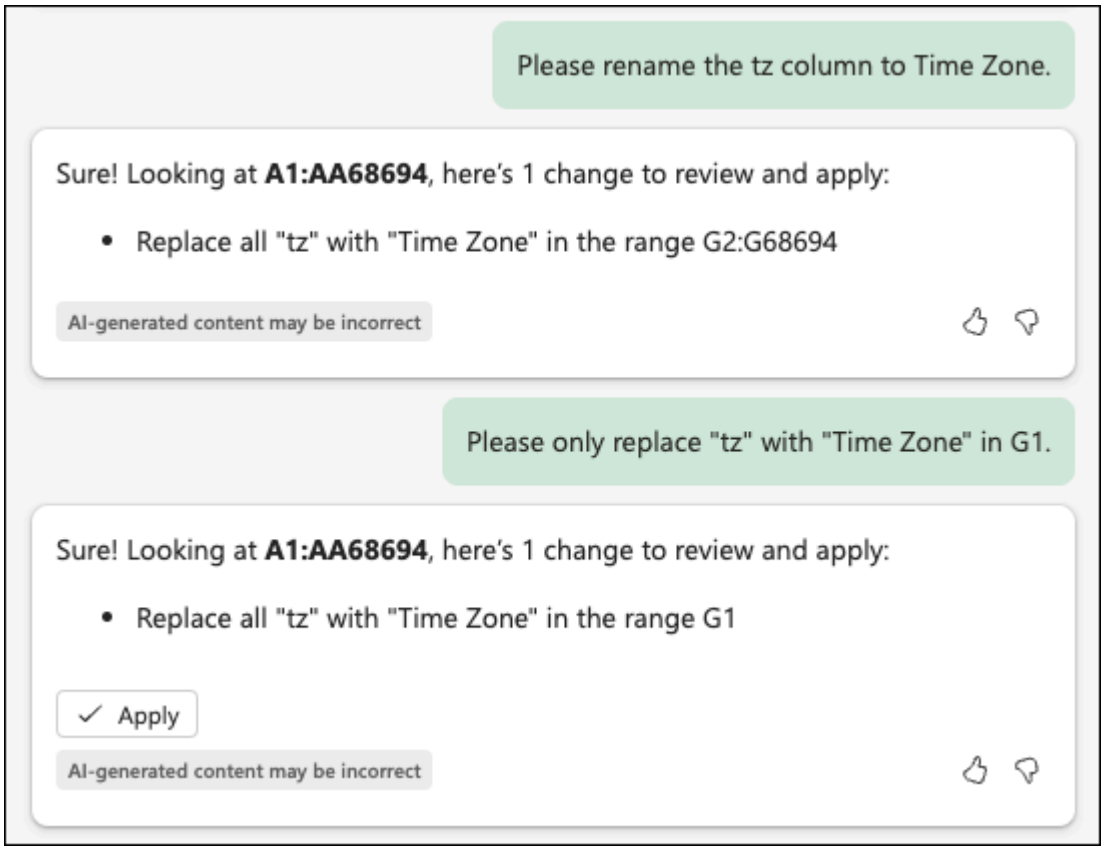


FIGURE 2-4: Clarifying where the replacement should take place.

This approach of replacing a single column header at a time is more time consuming than just doing it manually. But it occurred to me that I already have a simple comma-separated list of new column headers (from what Copilot in Edge generated) and perhaps Copilot in Edge would be able to use this list to correctly replace the headers.

I copied the list generated in Edge and prompted Copilot in Edge to use this list to replace the headers. However, Copilot came back with the message about there being an issue with renaming the column headers, as shown in Figure 2-5.

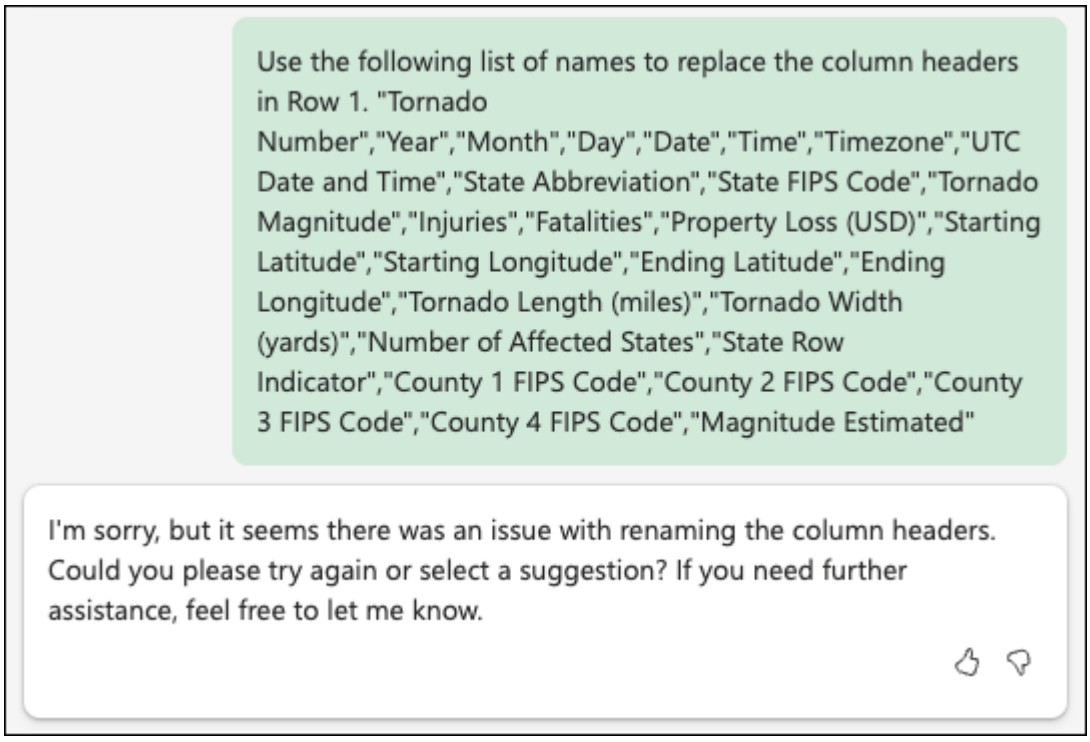


FIGURE 2-5: Copilot says there's an issue with renaming column headers.

Rather than argue with Copilot, you can use the following steps to manually replace the abbreviated headers with the new and descriptive headers:

1. **Copy the comma-separated list from Copilot in Edge and paste it into a text document.**
2. **Save the text document with the** .csv **extension.**
3. **Open the** .csv file **containing the headers in Excel.**
4. **Copy the new row and paste it into Row 1 of the tornado spreadsheet.**

Sometimes, the best approach to getting what you need to be done is to combine the use of Copilot with a manual approach. With some experience, you'll start to see where it's not worth it to try to get Copilot to do the last ten percent of a project when you could just do it yourself in a matter of seconds.

Formatting data

The next thing you may notice about the tornado dataset is that it's not sorted correctly, or it's sorted in a way that doesn't make sense. Looking at the first 20 rows, you'll see that the first records are from October,

November, and December 1950, and then the rows jump back to January through September 1950, as shown in Figure 2-6.

	A	B	C	D	E
1	Tornado Nu	Year	Month	Day	Date
2	192	1950	10	1	10/1/1950
3	193	1950	10	9	10/9/1950
4	195	1950	11	20	11/20/1950
5	196	1950	11	20	11/20/1950
6	197	1950	11	20	11/20/1950
7	194	1950	11	4	11/4/1950
8	198	1950	12	2	12/2/1950
9	199	1950	12	2	12/2/1950
10	200	1950	12	2	12/2/1950
11	201	1950	12	2	12/2/1950
12	4	1950	1	13	1/13/1950
13	5	1950	1	25	1/25/1950
14	6	1950	1	25	1/25/1950
15	7	1950	1	26	1/26/1950
16	1	1950	1	3	1/3/1950
17	2	1950	1	3	1/3/1950

FIGURE 2-6: A strange way to sort the year.

This pattern of the double-digit numbers coming first in the year holds true for all 68694 rows of data.

It's possible that this is a standard sorting scheme that I'm not aware of, but more likely, it's because the dates are currently sorted alphabetically rather than numerically. In alphabetical sorting, the number 10 comes before 2 because 1 comes before 2. I'd rather have all the data sorted chronologically.

If I wanted to sort by date, I could try clicking the column header. But, because the date is currently not in date format, it probably wouldn't work correctly. The next step in getting this data ready for analysis is to fix the data types.

Changing data types

Before you can sort by date, you have to make sure the Date column is in date format.

Other columns of the table that contain date information and numeric information should also be changed to the right data type. This seems like something Copilot might be able to help with.

I opened the Copilot sidebar and entered the following prompt:

Convert each column to an appropriate data type.

Copilot's response is shown in Figure 2-7.

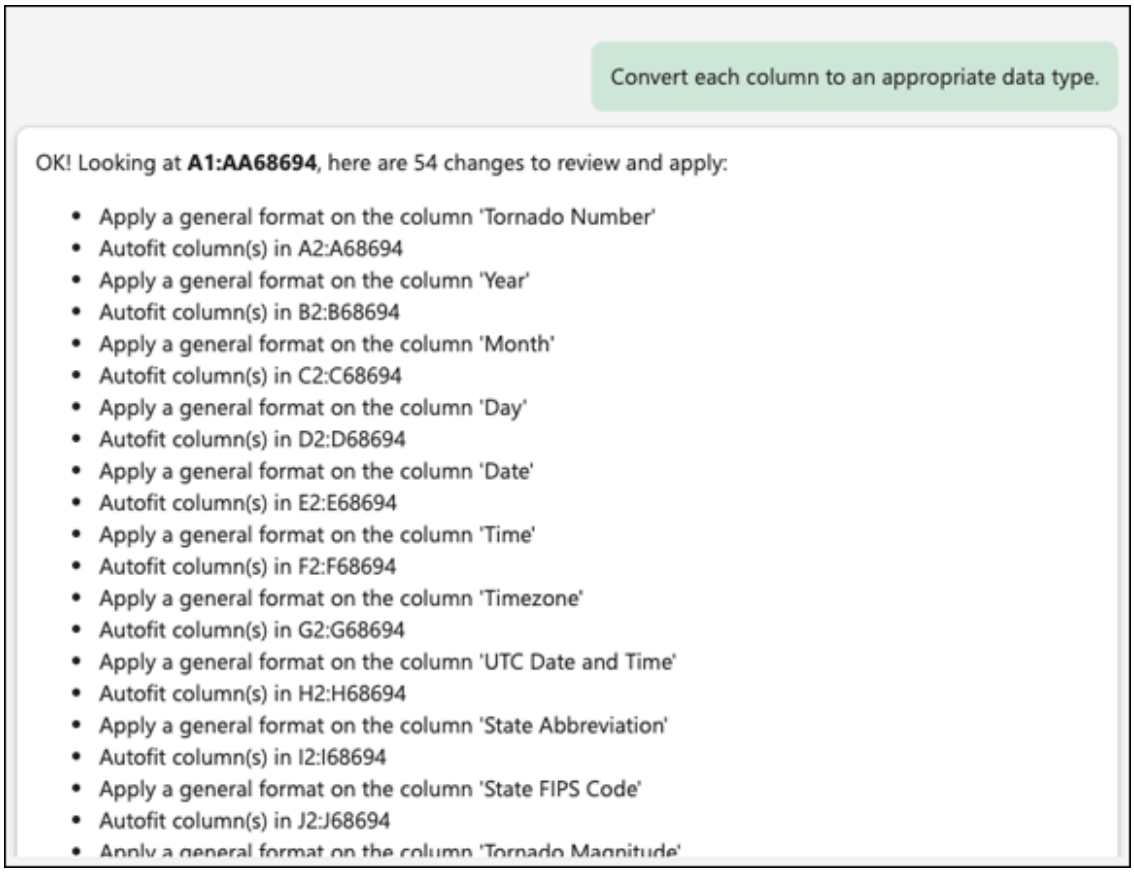


FIGURE 2-7: Copilot's proposed actions.

For each column, it suggested making the data type General and autofitting the columns, which I assume means making them wide enough so the data is visible.

Using the General data type for every column might work for most columns, but I have my doubts when it comes to the date columns, and I'd feel better if they used the date format. I prompted Copilot with the following:

Are there columns that should be changed to the Date data type?

Copilot replied that yes, certain columns would be more appropriately set to the Date data type, and then it listed those and asked me if I would like it to make these changes. I said yes. Copilot came back with a list of its proposed changes, which I accepted, and it made correctly.

I did the same thing to change the columns that should be stored as numbers to appropriate data types.

Sorting data

Now that the dates are correctly formatted, you can ask Copilot to sort the spreadsheet. The tornados dataset has a column with the date and time of each tornado. This date and time are standardized using UTC, so it's a perfect candidate for sorting.

I used the following prompt to ask Copilot to take care of it:

> *Sort the data on the UTC Date and Time column, in ascending order.*

Copilot showed me how it interpreted my prompt and provided an Apply button, as shown in Figure 2-8.

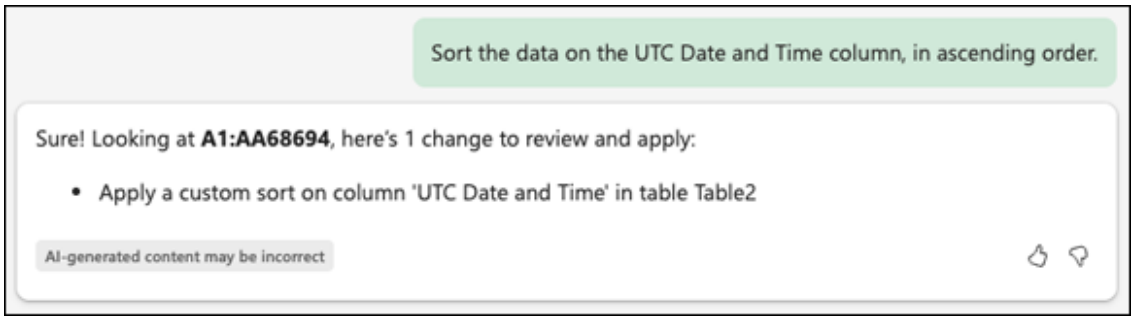


FIGURE 2-8: Copilot won't change anything without your approval.

I clicked the Apply button, and Copilot made the change correctly. It showed me the message in Figure 2-9.



FIGURE 2-9: Copilot's completion message and Undo button.

Cleaning data

When you have a subscription to Microsoft 365 Copilot or Copilot Pro, you can use the Clean Data tool to look for inconsistencies in a spreadsheet. To access the Clean Data tool, click on the Data menu on the ribbon. The Data tab has a section called Data Tools, and Clean Data appears near the middle of it, as shown in Figure 2-10.

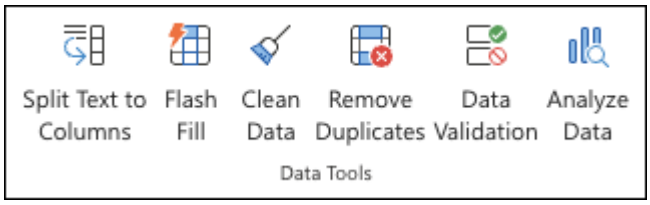


FIGURE 2-10: The data tools in Excel.

Clicking the Clean Data icon will cause the Clean Data with Copilot sidebar to open and display any suggestions that Copilot has for cleaning your data. At this point, however, Copilot didn't have any suggestions, as shown in Figure 2-11.

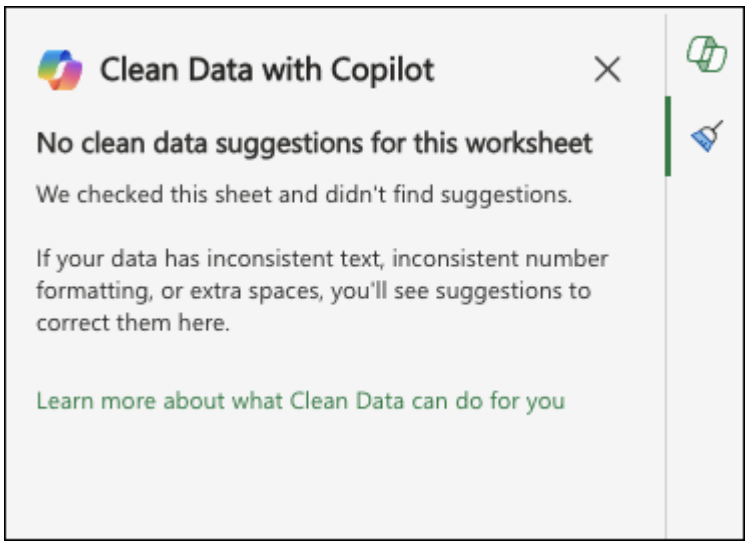


FIGURE 2-11: Copilot thinks the data is clean.

The next step in preparing the data is to look for any data that's obviously not correct and should be discarded. One way to find bad data is to look for outliers. *Outliers* are values that seem to be very different than the other values in a column. For example, in this dataset, tornado lengths generally range from 0.1 miles to around 70 miles. I don't know much

about tornadoes, but if there was a row with a tornado length of 700 miles, it would probably represent a data entry error.

You can have Copilot look for outliers using the following prompt:

Are there any outliers in my data?

When I asked this question, Copilot thought about it for a while but then responded that it couldn't determine if there are any outliers. This may have been because there's so much data, and it's treating every column in the dataset as equally important. As people, we know that, for example, the column containing state abbreviations is unlikely to have outliers, and it wouldn't matter if it did. Copilot in Excel doesn't seem to understand that.

I narrowed down the problem and asked again:

Are there any outliers in Property Loss (USD)?

Copilot was also unable to complete this. It may be that this dataset just has too much data for Copilot to look for outliers. I decided to try one more time and to ask if there are any outliers in starting latitudes. Since most tornados happen in the mid latitudes (between 30 and 50 degrees North and South), a tornado that occurs outside of the mid latitudes would be interesting at least, and possibly an outlier that should be looked into.

Are there any outliers in Starting Latitude?

Here again, Copilot wasn't able to complete the job and asked me to try again later. I decided to give it one more shot, but with fewer rows of data.

I made a copy of the dataset and asked Copilot to delete all but the first 1,000 rows of data. This phrasing of my request seems to have confused Copilot, because it only deleted rows 1,002 and the last row in the table. I manually deleted all but 1,000 rows of data and returned to my outlier prompts:

Are there any outliers in my data?

With far less data, Copilot was able to complete this request. I didn't find any significant outliers in this sample of the data, so I feel fairly confident that we're ready to move forward.

Now that you have some data, and it's cleaned up and the column headers are understandable, you're ready to try analyzing the data. Continue to the next section to see how Copilot can help.

Automating Data Analysis

Data analysis is all about finding useful connections between different pieces of data in a dataset. For example, analysis of your company's records of customer service calls might discover that an unusually high number of calls have to do with a particular product. This insight might be useful to the engineers working on that product.

Some insights that can be found in data are less useful, even though they may demonstrate strong relationships. For example, "discovering" that customer satisfaction with service calls is higher for calls in which the customer's problem was resolved would not be a particularly interesting insight.



REMEMBER Discovering insights into data, as with all Copilot uses, starts by asking a good question. For example, you might say, "I wonder if there's a certain day of the week when we get more calls than on other days."

Using Copilot for automated insights

An interesting and potentially useful feature of Copilot in Excel is its ability to automatically discover insights within your data. The idea of this feature is that Copilot will analyze your data to look for patterns and relationships that you may have missed, and it will even create a chart that shows the insight it found.

In reality, the result of having Copilot discover insights is often just funny. It doesn't seem to consider context or use knowledge of the outside world while suggesting insights. Copilot automatically creates a graph to show you what it finds. For example, Figure 2-12 shows a data insight Copilot gave me from a weather dataset I asked it to analyze, in which it points out that the moon phase (for example, full moon, new moon, first quarter, and last quarter) is a repeating pattern. This is quite literally one of the very first insights that humans ever had … right before discovering that the seasons are a repeating pattern that occurs in the same order every year.

Other similarly obvious insights Copilot gave me from this same weather dataset included the following:

- There is a strong correlation between rain and humidity.
- Visibility goes down after sunset.
- Wind gusts are higher when precipitation is higher.
- Snow and rain conditions have significantly higher precipitation.

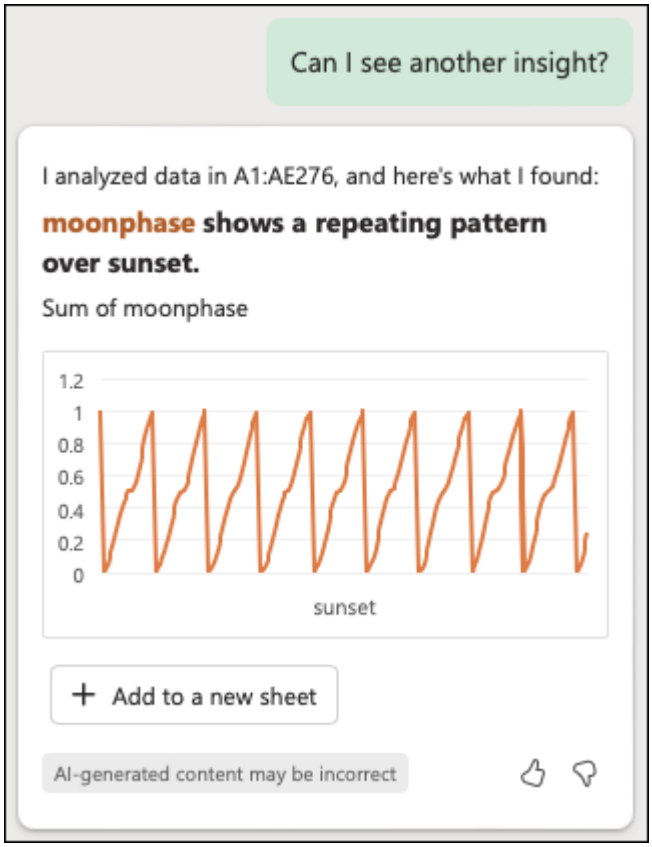


FIGURE 2-12: Copilot's insights are often not insightful.

My hope with the tornado dataset is that adding descriptive headers and giving Copilot more context about the data will encourage it to have better insights.

To ask Copilot to generate data insights, you can click the Show Data Insights link in the chat sidebar or simply type and submit the phrase *Show Data Insights* as a prompt.

After you ask it for insights, Copilot will analyze the data and come back with a single insight and a chart, as shown in Figure 2-13.

This first chart shows the number of fatalities due to tornadoes by year. The number is generally very low, but there are a couple of huge spikes. The chart isn't the greatest, however, because it doesn't show the years that correspond to the fatality numbers. I'm hoping Copilot can help improve the look and usefulness of the chart, too, but I cover that later in this chapter. To read more about using Copilot to create different kinds of charts and to modify charts, see the section of this chapter titled "Visualizing Data with Copilot."

To see the next insight, click the link under Copilot's latest response to ask for another insight, or just type your request for another insight into the prompt area and submit it.

You can continue to ask for more insights and Copilot will show up to five total insights without re-analyzing the data. After that, you can start over by asking for data insights, but it will likely respond with the same five.

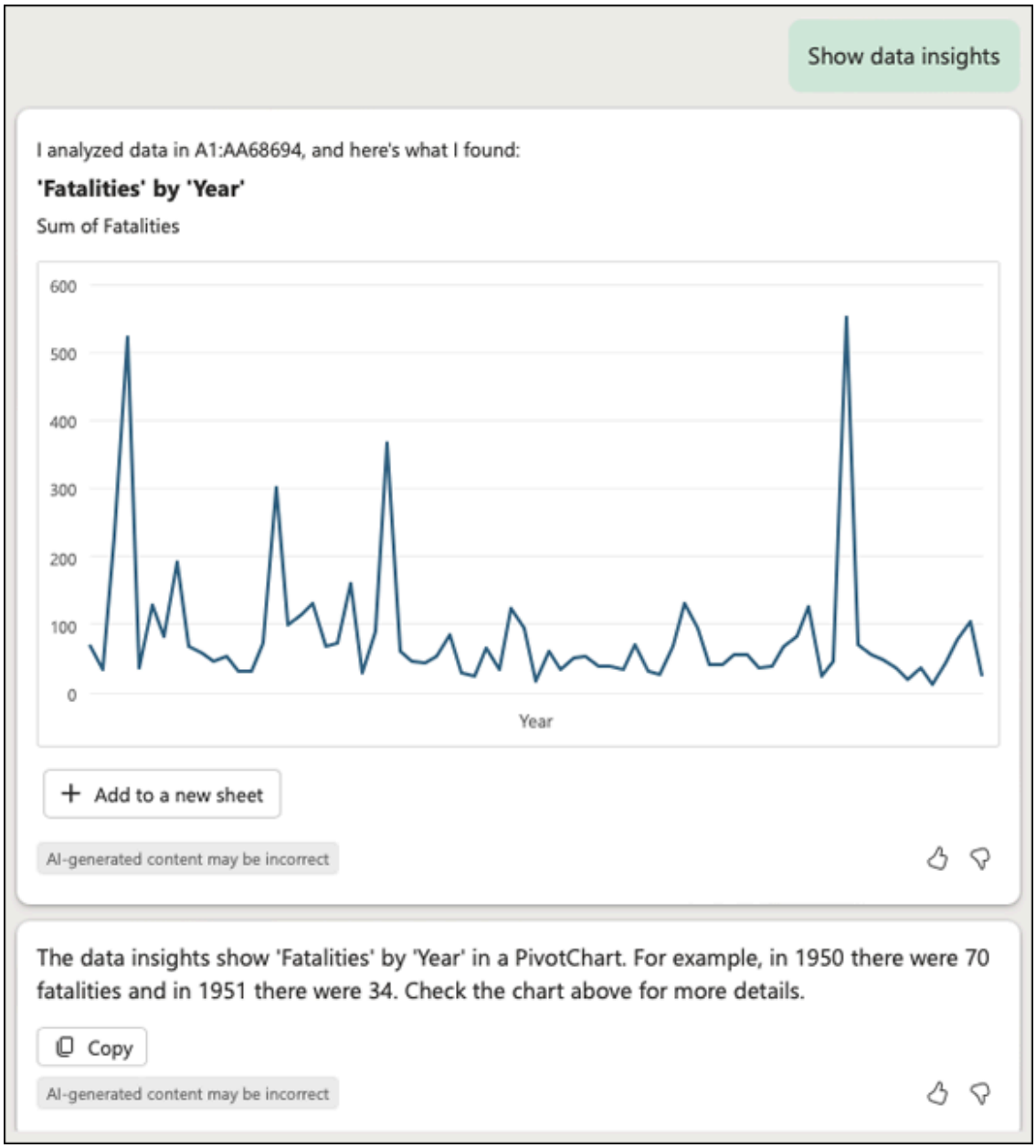


FIGURE 2-13: Copilot's first insight into the tornado dataset.

Unfortunately, in my prompting so far, Copilot hasn't been able to come up with many interesting insights on its own. At this point, you'll need to provide some guidance.

Asking for a specific analysis

Asking for a specific visualization produces much better results than asking Copilot to discover insights. For example, in Figure 2-14, I asked Copilot to show the relationship between tornado magnitude and property loss, and it successfully created a scatter chart. Oddly, after it generated a scatter chart it said that it couldn't create a scatter chart.

You learn more about prompting for specific charts and analysis in the section of this chapter titled "Visualizing Data with Copilot."

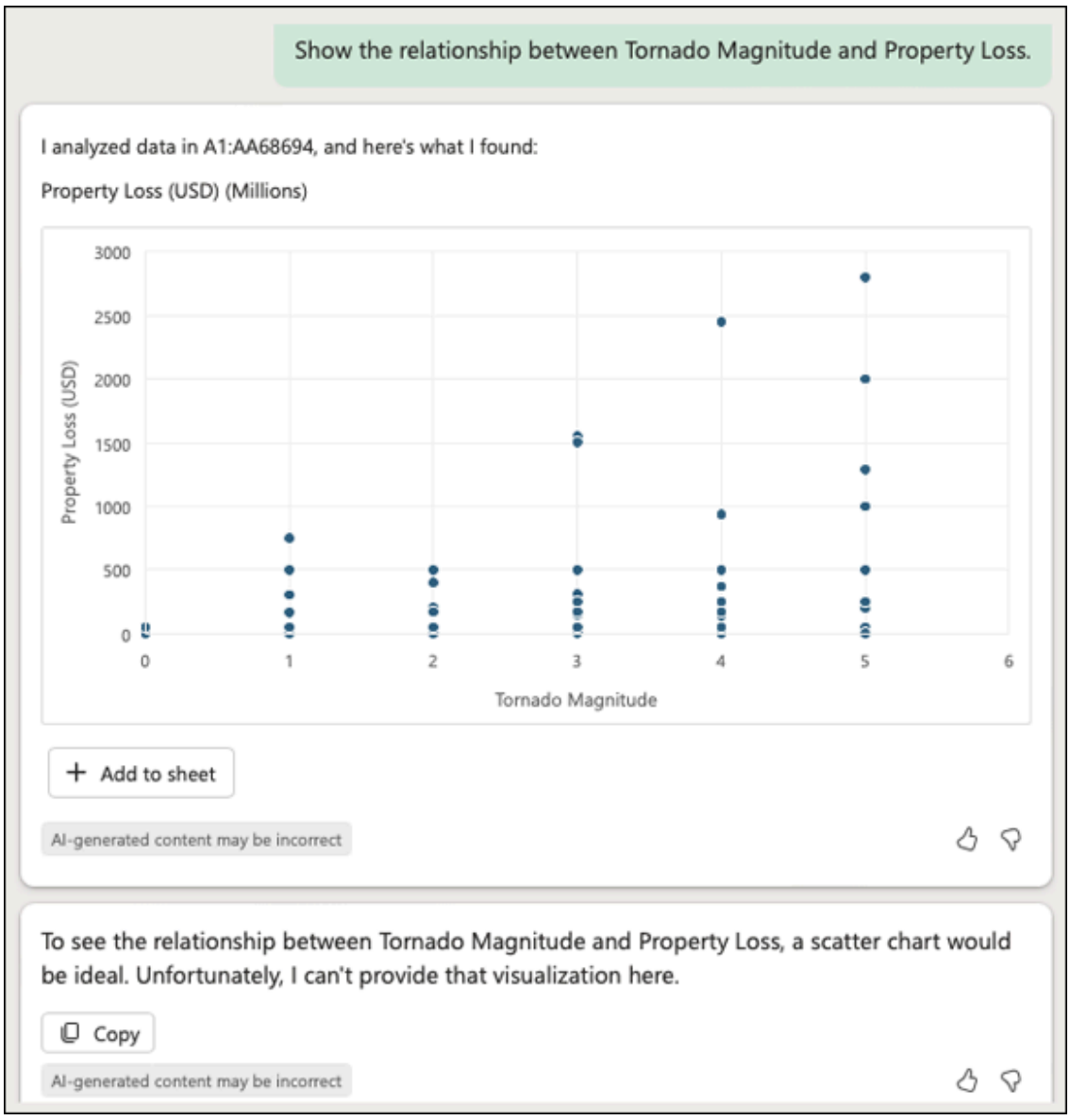


FIGURE 2-14: Copilot creates scatter charts but says it can't.

Creating Formulas with Copilot's Assistance

Formulas in Excel are what make it more than just a way to display raw data. With formulas, you can do many different kinds of operations, such as performing basic arithmetic, summarizing data, making conditional calculations, manipulating text, performing financial calculations, and analyzing data.

For those of us who don't use Excel on a regular basis, creating complex formulas can often be a matter of trial and error — or mostly error.

Copilot can help by automating the process of writing formulas and creating columns that calculate values based on other columns.

To try out this use of Copilot, I found a spreadsheet with a lot of potential for doing complex calculations. The data I'm going to be using contains weather data for the city where I live. The spreadsheet has detailed weather data for every day of a recent nine-month period. Figure 2-15 shows a small piece of the spreadsheet.

	A	B	C	D	E	F	G	H	I	J
1	datetime	tempmax	tempmin	temp	feelslikemax	feelslikemin	feelslike	dew	humidity	precip
2	2022-03-01	55.7	47.9	51.8	55.7	46.1	51.5	49.7	92.9	0.4
3	2022-03-02	50.2	45	47.4	50.2	40.9	46.4	46.9	97.6	0.36
4	2022-03-03	47.5	42.5	44.6	44.8	39.4	42.3	40.5	85.5	0.08
5	2022-03-04	48.1	40	43.9	45.4	37.9	42	39	83	0.02
6	2022-03-05	50.4	36.9	42.8	50.4	35	41.3	37.9	83.1	0
7	2022-03-06	49.8	39.8	43.9	48.7	38.7	42.5	38.4	81.2	0
8	2022-03-07	47.3	35.7	42.1	47.1	33	40.4	37.4	83.4	0
9	2022-03-08	47	44.5	45.8	45.2	41	43	42.8	89.1	0.14
10	2022-03-09	50.9	37.1	44.4	50.9	33.9	42.5	35.2	72.2	0
11	2022-03-10	49.9	31.8	40.6	48.2	30.8	39.9	28.8	64.6	0
12	2022-03-11	50.8	37.1	43.1	50.8	34.9	42	34.4	72.3	0
13	2022-03-12	52.8	34.1	43.9	52.8	30.3	41.2	37.5	79.1	0.28
14	2022-03-13	49.2	42.4	46	44.2	37.9	41.5	42.8	88.7	0.73
15	2022-03-14	49.8	45.4	46.9	48.5	42	44.6	42.9	86.4	0.5
16	2022-03-15	51.3	46.4	49	51.3	43.1	46	45.3	87	0.29
17	2022-03-16	50.8	40.7	45.5	50.8	38.1	44.5	41	84.5	0.08
18	2022-03-17	48.7	39.8	44.2	47.2	38.6	42.6	40.9	88.2	0.04
19	2022-03-18	52.9	45.1	48	52.9	44.3	46.9	44.7	88.6	0.11

FIGURE 2-15: A small piece of my weather data spreadsheet.

Formula creation and troubleshooting

The first thing I thought I'd like Copilot to create is a new column that uses a formula to keep a running total of the number of days with rain or snow. I submitted the following prompt:

> *Create a column that keeps a running total of the number of days with rain or snow.*

After a moment, Copilot gave me a proposed formula, complete with an explanation of what the new column will be. It also gave me the option to insert it into the spreadsheet, as shown in Figure 2-16.

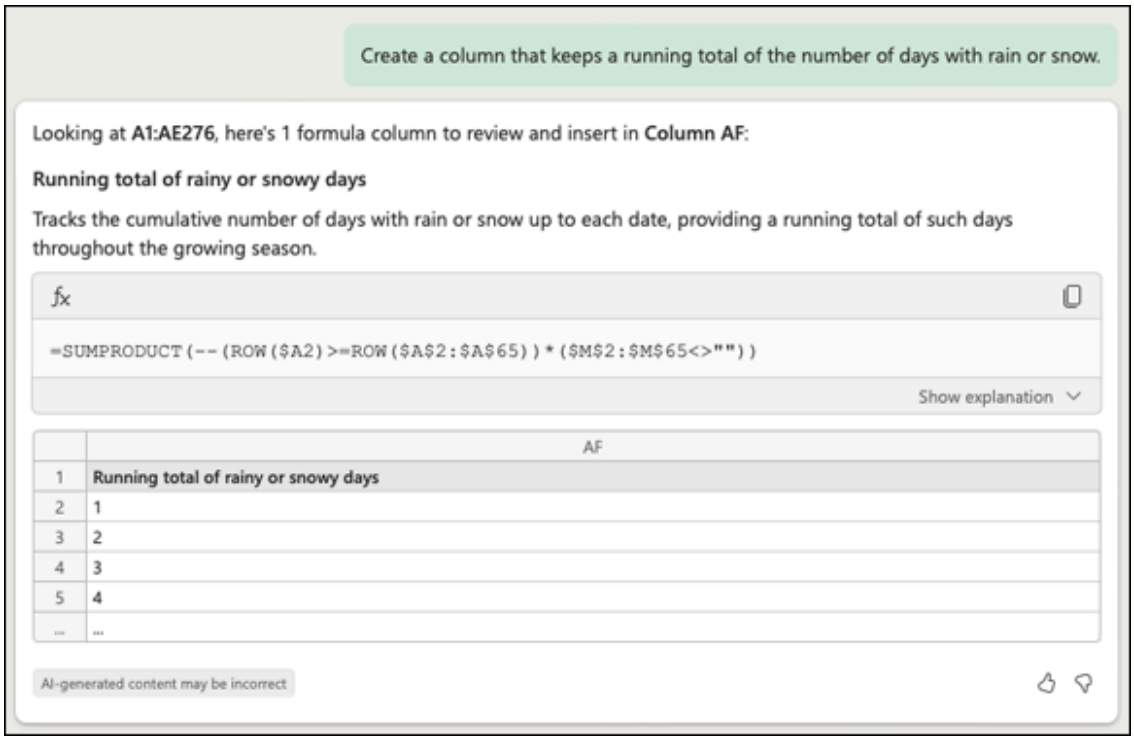


FIGURE 2-16: Copilot's proposed formula.

Honestly, I didn't know whether this formula would work correctly, and I couldn't tell just by looking at it. I clicked the button to insert the new column so I could investigate it thoroughly.

Looking at the results, it seemed to me like it was counting days with precipitation correctly, but I couldn't easily figure out how it was doing it. Here's the formula Copilot came up with:

```
=SUMPRODUCT(--(ROW($A2)>=ROW($A$2:$A$65))*
($M$2:$M$65<>""))
```

Whatever it's doing here, it puzzled me why the highest number row it's looking at is 65 when the spreadsheet has 276 rows, but I ignored that and looked at it some more. I struggled with understanding it for a couple of minutes, then asked Copilot how it worked, using this prompt:

> *Please explain the formula you used to create the running total of rainy or snowy days.*

The response Copilot gave is shown in Figure 2-17.

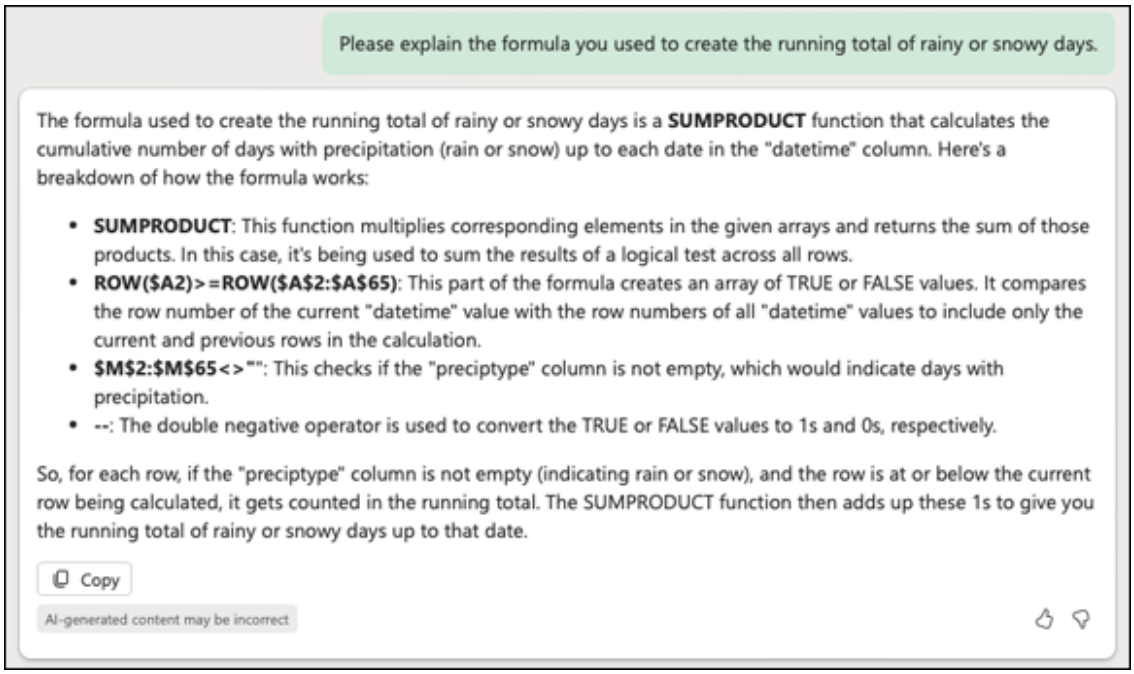


FIGURE 2-17: Copilot's explanation of its formula.

Copilot's approach to calculating the running total isn't the way I would have thought of doing it, but I am, admittedly, a dummy when it comes to Excel formulas.

Convinced that Copilot was at least on the right track, I did a manual check of the results from its formula by scrolling through the spreadsheet and counting. The formula that Copilot wrote stops updating after row 65. I changed the upper limit in the formula to the last row of data, which fixed the problem.



WARNING I've said it before, and I'll say it again: Don't blindly trust anything generative AI says or does for you. It can and will be wrong. Its ability to be creative and random is great for some tasks, but incorrectly analyzing data can have real-world consequences. Always double-check its work.

Advanced formula techniques

Copilot can potentially help with many data analysis tasks by creating formulas. The more you know about Excel, the more useful Copilot in Excel

is — especially when it comes to more advanced formulas. Here are some additional tasks Copilot can help you with:

- **Splitting and combining columns.** It's quite common to have a name column that you want to split into separate first name and last name columns. Although Copilot will likely get tripped up on names that don't stick to a standard first name last name format (such as people who use a middle name, or multi-word names), asking Copilot to do the initial work and then fixing up places where it makes mistakes is easier than doing this work manually.
- **Date and time calculations.** Want to calculate the difference between dates or add a specific number of days to a date? Copilot might be able to help here, too. However, make sure to watch out for unusual cases, such as leap years, that Copilot may not account for.
- **Text functions.** Text-manipulation tasks include tasks like extracting specific parts of a string, converting state or country name abbreviations to full names, and concatenating (combining) words from different cells. The key here is to know when you're better off simply using Excel's standard Find and Replace functionality.
- **Conditional formulas.** Conditional formulas apply different calculations based on conditions. For example, you might ask Copilot to create a formula that reports the wind speed only on days when it's rainy.

Visualizing Data with Copilot

Visualizing is what data analysts call the process of creating charts, graphs, and any graphical representation of information and data. Data visualization makes data accessible and can help people understand trends, outliers, and patterns in the data.

To demonstrate Copilot's ability to create data visualizations, I've chosen another dataset from Kaggle, titled "700 Classic Disco Tracks (with Spotify Data)." I chose this dataset because the appeal of disco music has always puzzled me, and I want to try to understand it better. Also, it's a welcome change of mood from the scary tornado data from earlier in this chapter.

So come along and ride on a fantastic voyage into disco and Copilot magic.

The disco dataset is a great example of creating structured data (rows and columns) from unstructured data (disco songs). It contains basic informa-

tion about each track, such as the title, artist, year, and duration, as well as the numbers that Spotify uses to categorize songs and create playlists. Each song on Spotify has a unique combination of scores from 0 to 1 on the scale of "Danceability," "Energy," "Speechiness, "Acousticness," "Instrumentalness," and "Liveness."

If you want to find out more about the dataset and experiment with it yourself, you can download it from `https://www.kaggle.com/datasets/thebumpkin/700-classic-disco-tracks-with-spotify-data/data`.

Opening and cleaning data in Excel

After downloading and extracting the compressed file from Kaggle, I opened it in Excel in Microsoft 365. Before you can edit the data or use Copilot with it, the file must first be converted from CSV data to the latest Excel format. To do this, use the drop-down menu in the upper right of the Excel program to switch the mode from Viewing to Editing. Excel will notify you that the file needs to be converted to the latest Excel file format, and you can agree to make that change.

Next, I converted the data to a table using the Format as Table tool that I previously used to convert the tornado data into a table.

After converting the file, Copilot notified me that it had three suggestions for cleaning the data. Its suggestions were super bad, however. It suggested fixing several song titles that it viewed as inconsistent. I know that Boogie Oogie Oogie doesn't mean anything, but this isn't the sort of change you should make when cleaning data. I ignored the suggestions.

I was curious about other suggestions that Copilot might have for cleaning the data, so I tried using the Clean Data tool again. Copilot didn't have any suggestions.

Creating charts and graphs

You've already seen how to ask Copilot to create charts using patterns it finds by prompting Copilot with "show data insights." I didn't have high

hopes for this one, but I thought I'd take a chance and try it out.

The first insight showed a mostly straight line to demonstrate the relationship between loudness and year. Oddly, it showed a large dip in loudness in 2018. I suspect this might be just because there wasn't a lot of disco music in 2018. Upon checking the raw data, it did turn out that there was only one song listed for 2018.

Another insight showed that nearly all disco songs have a 4/4 time signature. This isn't surprising — I assume there aren't a lot of disco dance moves that work well with waltzes.

The insights I'm most interested in are what makes a disco song popular. The dataset has a column named Popularity that assigns a number to each song. The higher the number, the more popular the song. I used the following prompt to attempt to find out what correlates with popularity:

Create a PivotTable of Popularity by Danceability.

(By the way, a PivotTable in Excel is a data summarization tool that allows you to reorganize and analyze large datasets dynamically.)

Unfortunately, Copilot said it was unable to create this PivotTable, so I tried something else:

Add a PivotTable of Popularity by Year.

This time, Copilot created the chart showing the average popularity of songs by year, as shown in Figure 2-18. This chart is largely useless, however, because the years that have the fewest represented songs create the highest and lowest points on the chart.

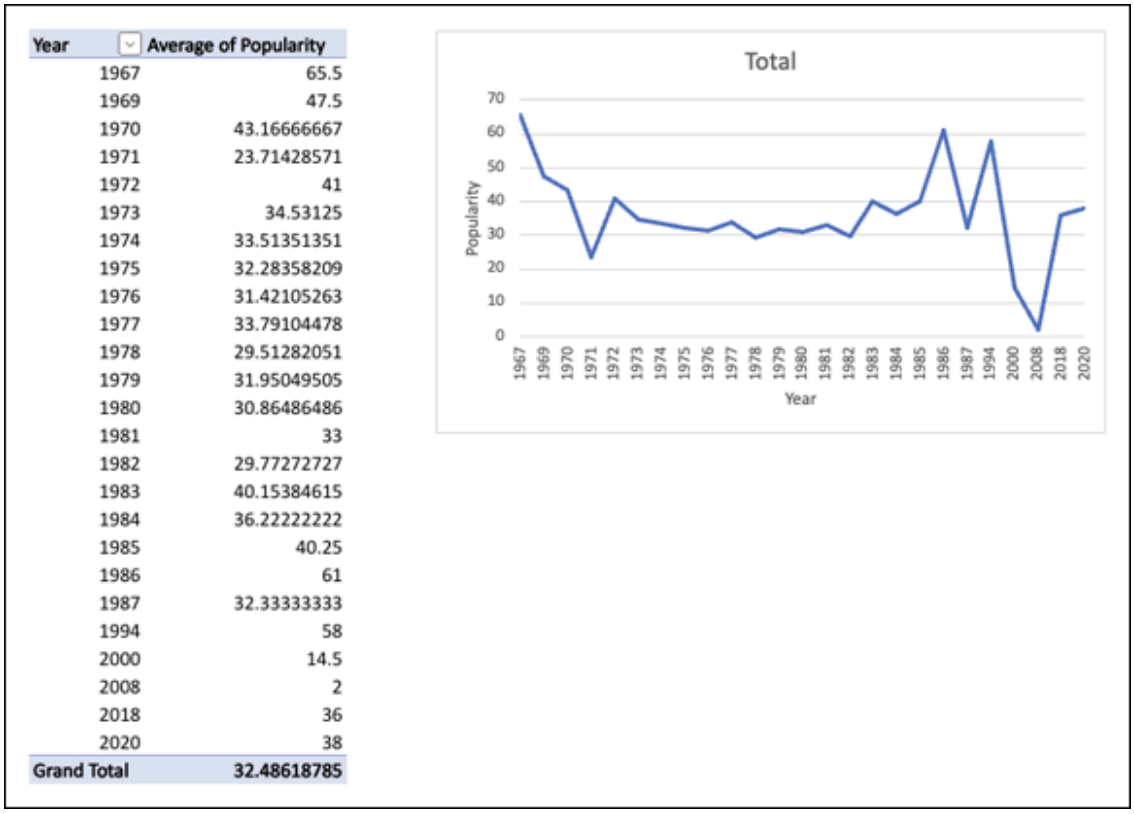
My next request resulted in something more interesting. Here's the prompt I used:

Add a chart of the highest popularity for each year.

Copilot generated the chart correctly, followed by a confusing and inaccurate message saying that it couldn't create the chart and also that it creat-

ed the chart, as shown in Figure 2-19.

After it generates a chart or report in the chat sidebar, Copilot gives you the option to add the generated content to a new sheet, which I did in this case.



Year	Average of Popularity
1967	65.5
1969	47.5
1970	43.16666667
1971	23.71428571
1972	41
1973	34.53125
1974	33.51351351
1975	32.28358209
1976	31.42105263
1977	33.79104478
1978	29.51282051
1979	31.95049505
1980	30.86486486
1981	33
1982	29.77272727
1983	40.15384615
1984	36.22222222
1985	40.25
1986	61
1987	32.33333333
1994	58
2000	14.5
2008	2
2018	36
2020	38
Grand Total	32.48618785

FIGURE 2-18: Average popularity by year.

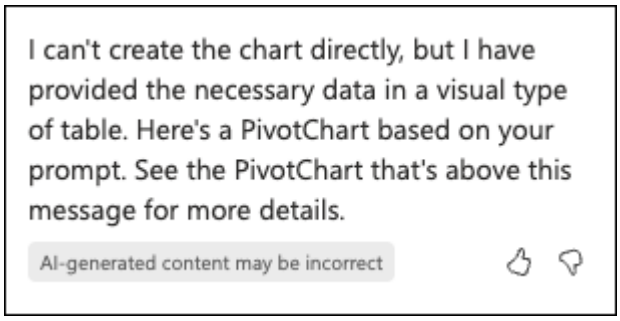


FIGURE 2-19: Copilot sometimes just says all the things.

The resulting chart, shown in Figure 2-20, shows that disco reached its peak popularity in 1976 and has mostly been dropping ever since.

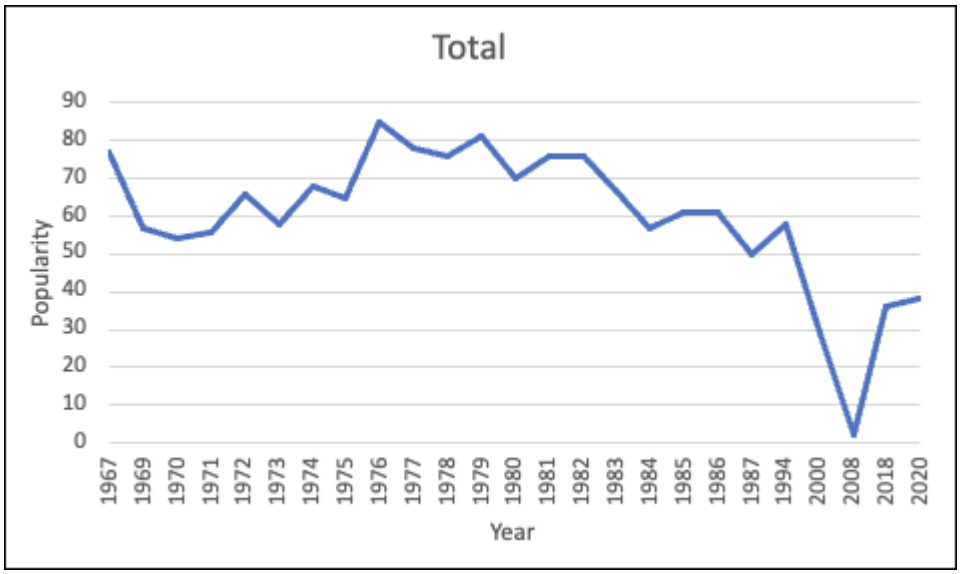


FIGURE 2-20: Max of popularity by year.

Customizing charts

Copilot can make certain kinds of changes to charts after they're created. To experiment with customizing charts, I created a new chart using the following prompt:

> *Chart the most danceable track by year.*

The resulting chart shows, as you'd expect, that the maximumly danceable track for each year is always ultra-high and sometimes appears to get close to 100 percent danceable, which might just be the point at which you can't stop dancin'. The chart Copilot created is shown in Figure 2-21.

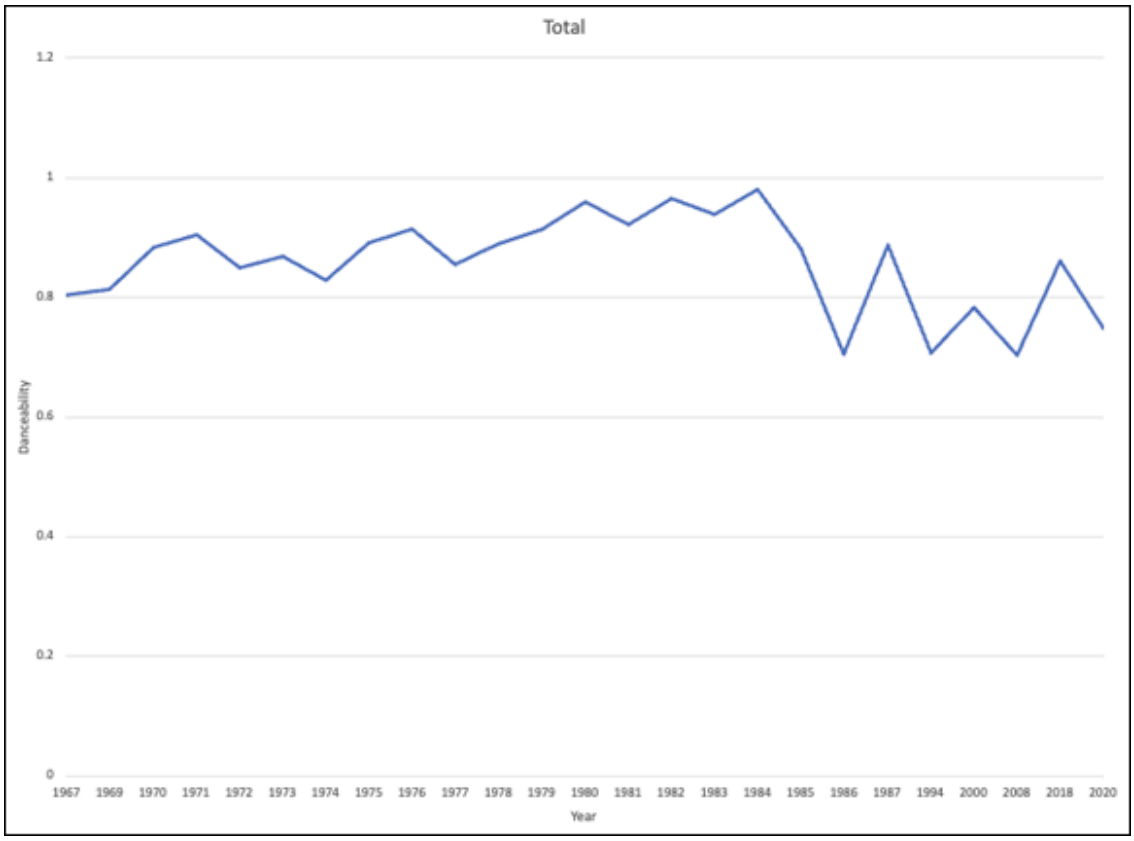


FIGURE 2-21: The initial danceability chart created by Copilot.

It would be useful to label some of the data points on the line chart, so I gave Copilot the following prompt (as well as several variations):

> *Add labels to the data points on the chart.*

Copilot repeatedly claimed not to be able to add data labels to the chart, so I switched my tactic and gave it the following prompt:

> *How can I add data labels to the line chart?*



WARNING Copilot gave me step-by-step instructions for adding the data labels myself, which is when I discovered an important limitation to the ability of Copilot to create and work with charts: *If you want to add a chart to a new sheet, you have to do it in the next prompt after the chart is created. If you don't, there's no way to go back in the chat and do it.* I had to start a new chat and ask Copilot to create the chart again. After that, I was able to add it to a new sheet.

I followed Copilot's instructions for adding data labels, and they worked fine, but what I wanted to do was more complex than the question I had initially asked, so I asked a different question:

> *How can I label the points in the line chart with the track titles?*

Copilot's response to this question looked good at first glance, but when I tried to follow the steps, the options and links Copilot said to use didn't exist. I suspect that the instructions Copilot gave me were for a different version of Excel. I gave up on using Copilot to customize the chart and searched the Microsoft Support website (`https://answers.microsoft.com/en-us/msoffice/forum/msoffice_excel`) instead.

Considering Copilot's Limitations in Excel

Copilot in Excel has some really interesting capabilities, and it's sometimes surprising what it's capable of. However, at this point, there are frustrating limitations to what it can do, and it's often wrong when it tells you what it is and isn't capable of doing. Sometimes submitting the same prompt more than once, or starting a new chat and submitting the same prompt will completely change the response.



TIP Starting a new chat seems to be the best way to improve the chances that a certain prompt will be successful.

Another major limitation of Copilot in Excel is its lack of knowledge of anything outside of your spreadsheet. This leads to data insights and suggested prompts that don't make sense.

I'm confident that Copilot in Excel will improve in the coming weeks and months, and the version you're using now may be markedly better. If it isn't, we still (thankfully) have traditional search engines, FAQs, and human Excel experts.